Moinuddin Quazi C-54

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Q1] What are the various security issues involved in mobile Computing?

→ Various security & issues involved in mobile computing are-

1) Device theft & loss.

- Mobile devices are highly portable & can be easily lost or stolen. Unauthorized access to sensitive data if the device isn't properly secured.
- Mitigation - Use device encryption remote wipe capabilities strong passwords and biometric locks.

2) Insecure networks-

- Mobile devices frequently connect to public or unsecured wifi networks.
- Exposure to man-in the middle attacks, eavesdropping & session hijacking.
- Mitigation :- employ VPNs, avoid sensitive transactions on open networks & use network security protocol.

3) Malware & malicious applications-

- Mobile ~~plat~~ platforms, especially those with open ecosystem like Android, are susceptible to software infections.
- Malware can steal data, track user behaviour or hijack device functionality.
- Mitigation:- Install apps only from trusted sources use mobile security software & keep the device's OS updated.

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4) Operating System & Software vulnerabilities:
· Mobile operating systems & third-party apps may have updated vulnerabilities. Attackers can exploit these vulnerabilities to unauthorized access or control over the device.
· Mitigation: Regularly update the OS & the application & monitor for security advisories.

5) Data leakage & storage issue
· Sensitive information stored on the device might be inadequately protected. Data leakage through insecure storage, encrypted backups or poorly managed permissions.
· Mitigation:- Use strong encryption for data storage, secure cloud ~~f~~ backups & enforce strict data access policies.

6) Authentication and authorized weakness:
· Weak passwords or inadequate authentication mechanism can be exploited. Unauthorized access if an attacker bypass or guess login credentials.
· Mitigation - Implement multi factor authentication (MFA) use robust password policies & consider biometric ~~s~~ security measures.

7) Wireless protocol vulnerabilities
· Technologies like Wi-Fi, bluetooth, NFC & even cellular communications can be targeted. Unauthorized data access or device manipulation through insecure wireless protocol.
· Mitigation - Disable wireless services when not in use keep the device updated & use secure

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passing protection.

2) write short note on Antennae.
→ · Antenna is a device that converts electromagnetic radiation in space to electrical currents in conductors or vice versa, depending on whether it is being used for recieving or for transmitting respectively.

The radiation pattern of an antenna describes the relative strength of the radiated field in various direction from the antenna, at a constant distance

An antenna can be used either as a transmitting antenna or recieving antenna.
· A transmitting antenna is one, which converts electrical signals into electromagnetic waves & radiates them.
· A recieving antenna is one which converts electromagnetic waves from the recieved beam into electrical signals.

In two way communication, the same antenna can be used both transmission & reception

Types of antennas:-
1) wire antennas.
· They can be found in vehicles, ships, aircraft, etc
· wire antenna come in different shapes & sizes like straight wire, loop & helix.
· Radiation pattern = Omni-directional pattern.

i) Omni directional antennas:

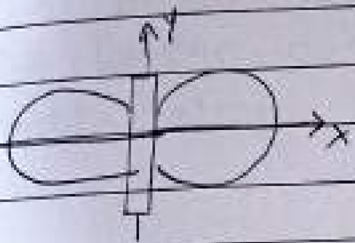
- These omni directional antennas are real antennas. The dipole radiation pattern is 360° there in the horizontal plane & approximately 75° in the vertical plane.
- It is also called the "non-directional" antenna because it does not favor any particular direction.
- Dipole antennas are said to have gain of 2.14 dB which is in comparison to an isotropic antenna. The higher the gain of antennas, the smaller the vertical beam width.
- It is used for broadcasting a signal to all points of the compass or when listening for signals from all points.

Dipole:-

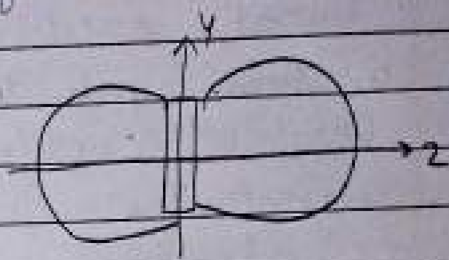
- The most commonly used antenna is Hertzian dipole.

The dipole consists of two collinear conductors of equal length, separated by a small feeding gap.
- The length of dipole is half the wavelength λ of the signal.

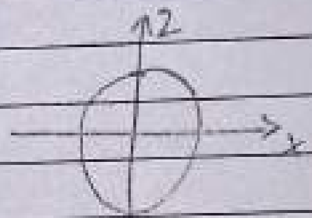
· dipole has uniform or omni directional radiation pattern in one plane & figure eight pattern in other two planes.
· This type of antenna are used in mountain valley, etc.
· It is simple antenna. it is difficult to mount of a roof top of vehicle.



sideview (xy plane)

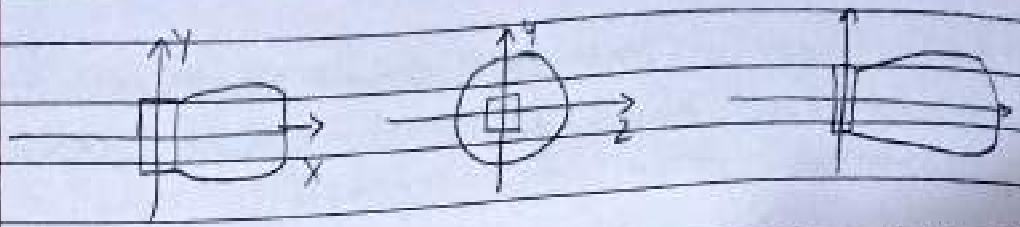


sideview (zy plane).



Top view (xz plane)

2) Directional antennas-
· A directional antennas or beam antenna which radiate or receives greater power in specific directions.
· This allows increased performance & reduced interference from unwanted sources.
· Directional antennas must be aimed in the direction of the transmitter or receiver.
· example:- parabolic & yagi antenna



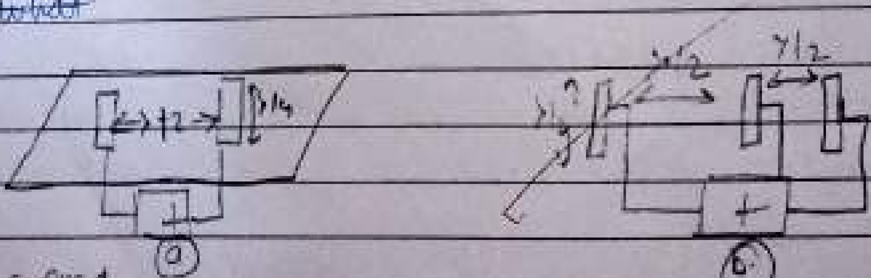
sideview(XY)   Sideview (YZ plane)   Top view(XZ)

3) Sectorized antennas:
· Several directional antennas can be combined on a single pole to construct a sectorized antenna.
· They are widely used in cellular telephony infrastructure.

Top view, 3 sector   Top view 6 sector

4) Antennas array:
· An antenna array is configuration of multiple antennas (elements) arranged to achieve a given radiation pattern.
· Multiple antennas allow different diversity scheme to improve the quality & reliability of a wireless.

5) ~~what~~



(a)   (b)

ground plane

3] What are various issues in signal propagation?
→ Signal propagation key several factors that impact the strength, quality & ~~the~~ reliability of transmitted signals.
These issues can cause signal degradation, interface, & loss of data. Below are key issues in signal propagation.

1. Attenuation :-
As signal travel, their strength decreases due to distance, obstacles, & medium resistance. Higher frequencies experience greater attenuation, making long-range communication challenging.

2. Distortion -
Different frequency component of a signal may attenuate at different rate, altering the original signal shape. This affects data transmission accuracy.

3. Dispersion -
The spreading of a signal pulse over time leads to inter-symbol interfence (ISI), where overlapping signals causes errors in data reception.

4. Noise -
External & internal disturbances affect signal quality, ~~including~~
- Thermal noise (due to electron movement in circuits)
- Crosstalk (interference from nearby wires).
- Impulse noise (sudden spikes from electromagnetic interference).

5. Multipath propagation:-
Signals may take multiple paths due to reflection, refraction, & diffraction, leading to fading & ISI, which degrade communication quality.

6. Reflection, Refraction, Diffraction & Scattering:-
These phenomena occur when signals interact with surfaces and objects, altering their path & reducing signal clarity.

7. Shadowing -
Large obstacles like buildings & hills block signals creating weak reception areas known as "dead-zones"

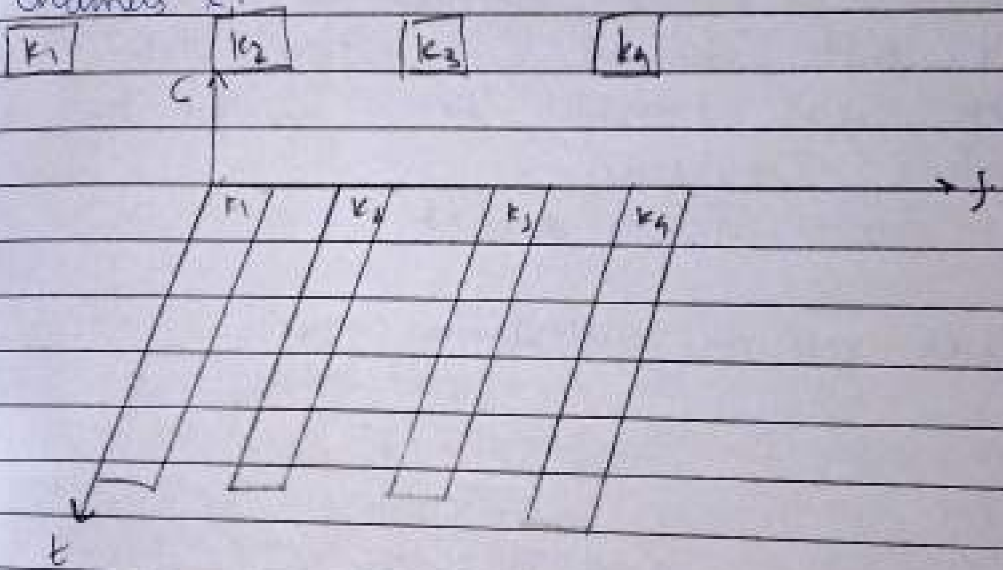
8. Doppler effect:
When senders or receiver is moving, the frequency of the signal changes, leading to comm^n delays or data errors.

9. Interference -
Overlapping signals from nearby channels cause co-channel & adjacent-channel interference

10. Signal propagation Ranges -
Transmission, detection & interference ranges define signal effectiveness.

Addressing these issues through error correction, power control, & adaptive techniques enhances wireless comm^n reliability.

Q) Discuss Multiplexing in wireless communication -
→ Multiplexing in wireless Communication:
Multiplexing is a technique used in wireless communication to allow multiple signals to share the same transmission medium effectively. This ensures optimal use of bandwidth & reduces interference while enabling simultaneous commⁿ between multiple users.
Types of Multiplexing:
1. Frequency Division Multiplexing (FDM):
Channels k_i



- The total bandwidth is divided into multiple non-overlapping frequency bands.
- Each user is assigned a unique frequency band for continuous transmission
eg. FM radio, analog TV transmission

2. Time Division Multiplexing (TDM)

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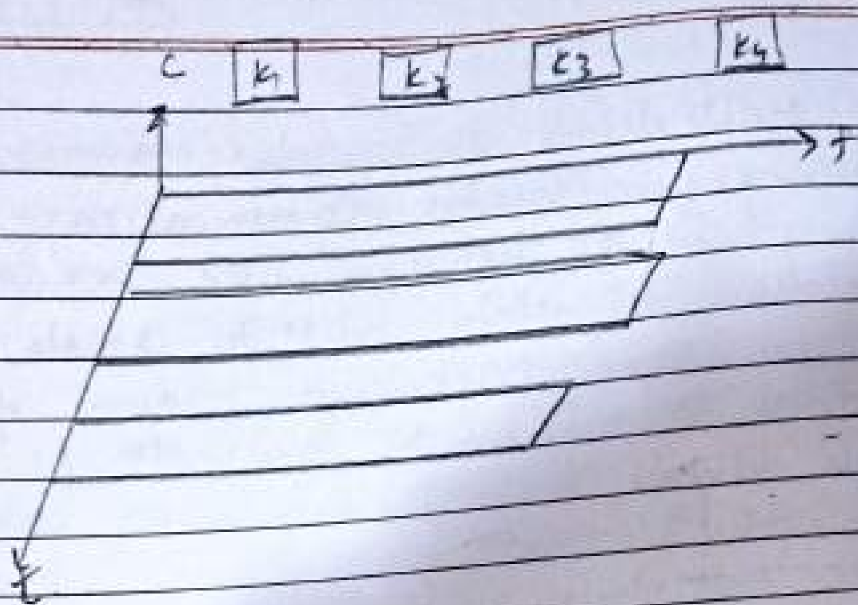
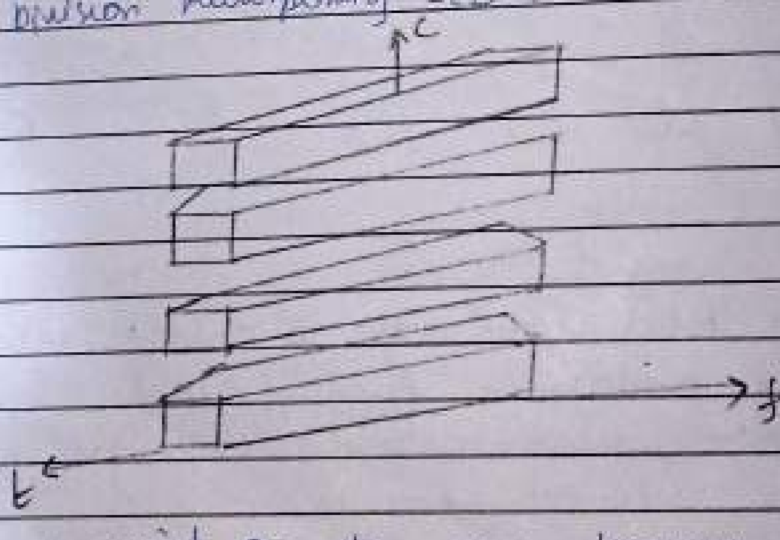


· The entire frequency spectrum is allocated to users but divided into time slots.
. Each user transmit in a specific time slots avoiding interference.
ex.: GSM mobile network.

3 Code Division multiplexing C(DM):

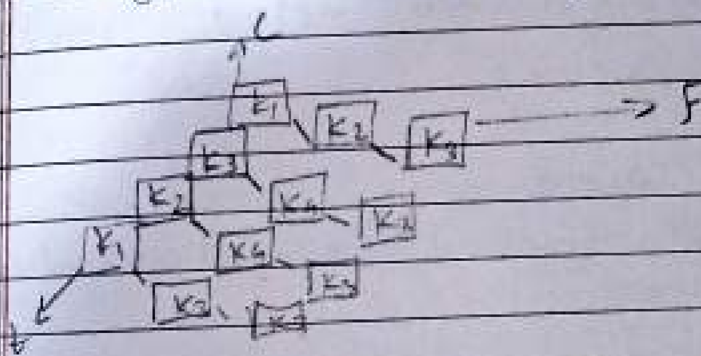


· All users transmit on the same frequency band simultaneously, but each is assigned a unique scale
. signals are seprated using orthogonal codes unique codes.
· signals are seprated using orthogonal codes to avoid interference

eg 3G cellular networks

4. Space Division multiplexing (SDM).
. Different users operate in seperate spatial locations using multiple antennas (MIMO tech).
. Used in cellular networks & satellite comm^n
eg. 5G MIMO technology.

5. Hybrid multiplexing (TDM / FDM).



. A combination of TDM & FDM to provide better efficiency & security
. eg. GSM networks use both TDM & FDM for improved performance

5) What are the main benefits of spread spectrum system? Explain spread spectrum in detail. How can DSSS system benefit from multipath propagation?
→ Benefits of spread spectrum system:
Spread spectrum tech^n spreads a signal ~~the~~ over a wide frequency range, making it more resistant to interference & improving security
The main benefits include:
1. Resistance to interference

2. Multipath resistence-
It mitigate multipath fading by spreading the signal over ~~of~~ different frequencies.

3. Security and Privacy-
The signal appear as noise, making them hard to detect or interupt.

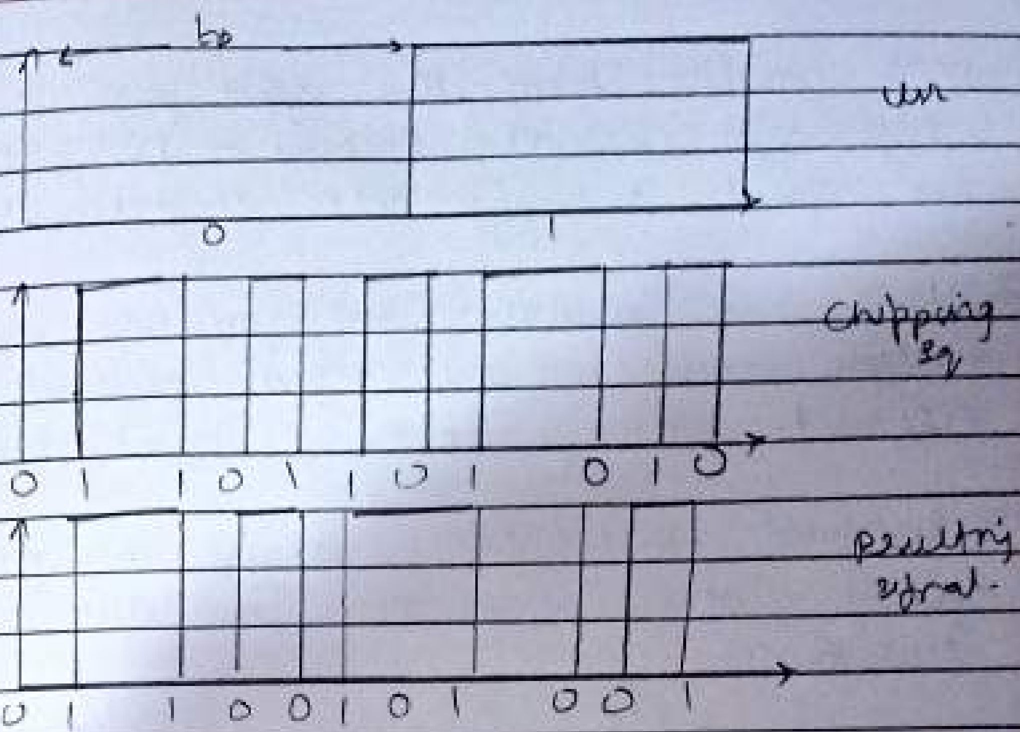
4. Resistence to Jaming-
Spread spectrum techniques are robust against intentional signal jamming.

5. Higher channel capacity -
Multiple users can transmit simultaneously using different codes

6. Improve signal quality -
Reduces distortion & ensures better signal reception

- Direct sequence Spread spectrum (DSSS).

- The direct sequence spread spectrum is a spread spectrum modulation technique primarily used to reduce overall signal interfence in telecomm^n.
- The DSSS modulation makes the transmitted signal wider in the bandwidth than the info^n bandwidth
- In DSSS, the message bits are modulated spreading ~~seq~~ sequencing process known as a ~~spreading sequence~~, chip.
It has a much shorter duration (DSSS) system take a user bit stream & perform as (XOR) with a chipping sequence as shown in figure



- The example shows that the result is either the sequence 0110101 or its complement 1001010.

- Advantages of DSSS:
1. less susceptible to noise & interference
2. Harder to detect, improving security.
3. Provides best protection against multipath signals.
4. Supports Code Division Multiple Access (CDMA): for multiple users.

- Applications of Direct Sequence Spread Spectrum (DSSS):
- Used in LAN technology
- Satellite communication technology
- Military & many other commercial application.
- It supports Code division multiple access

How DSSS benefits from multipath propagation:
Multipath propagation occurs when signals take multiple paths due to ~~m~~ reflections, refractions &

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· DSSS benefits from this effect by:
1. Improving signal recovery - DSSS spreads the signal, reducing inter-symbol interference (ISI)

2. Using Rake receivers - these receivers combine multiple copies of the signal received from different paths, improving signal quality.

3. Enhanced reliability - even if one path experiences fading, other signal copies can still be used for reconstruction.

4. Thus, DSSS provides better robustness against multipath fading, making it ideal for wireless & mobile communication systems.

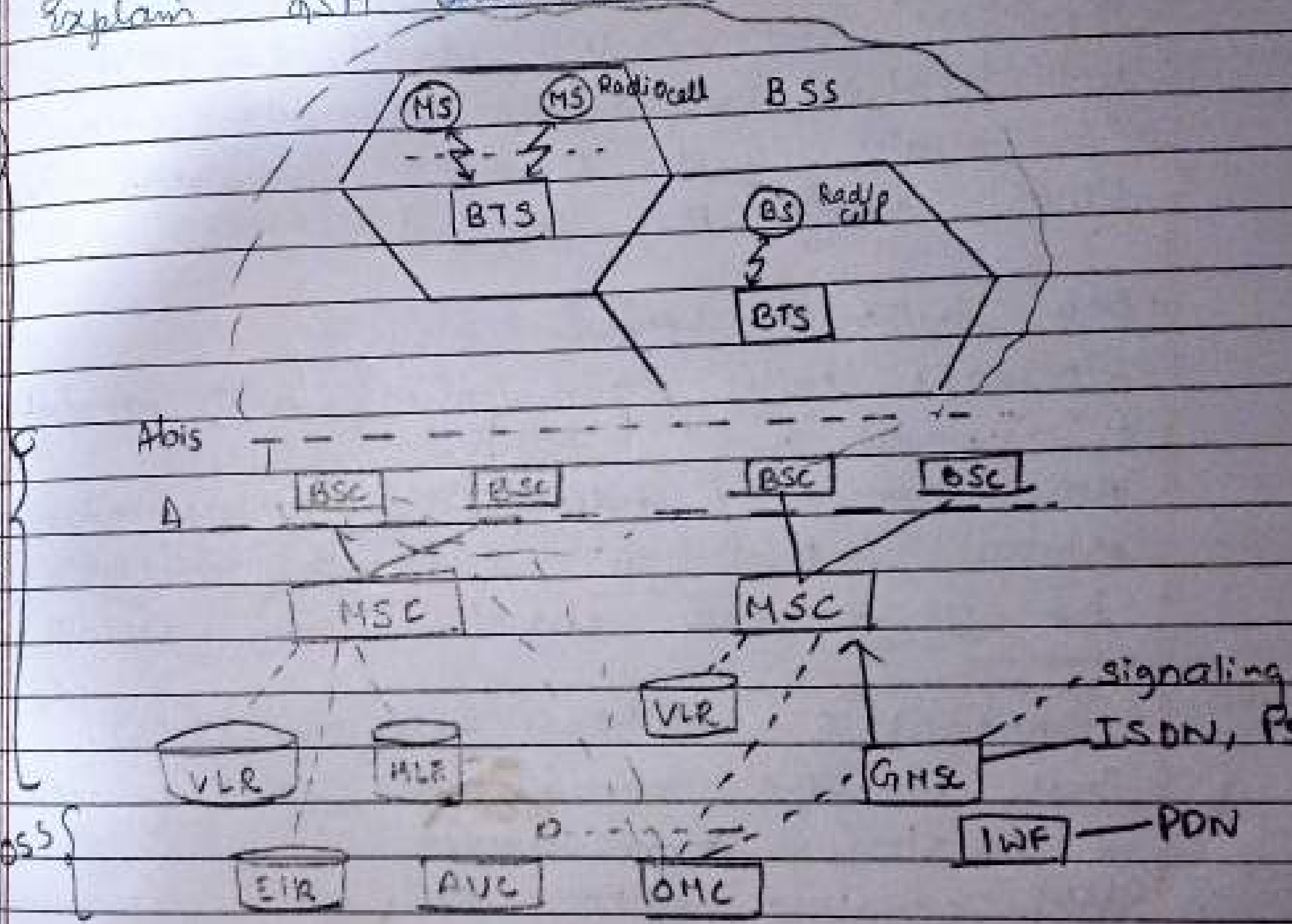
6) Compare various Telecommunication Generation.

→

	1G	2G	3G	4G	5G
Period	1980-1990	1990-2000	2000-2010	2010-2020	2020-2030.
Band width	150/900 MHZ	900 MHZ	100 MHZ	100 MHZ	1000x BW per unit area
frequency	Analog signal	1.8 GHZ (digital)	1.6-2.0 GHz	2-8 GHz	3-300 GHz
Data rate	2Kbps	64 Kbps	144 - 2 Mbps	1 Mbps - 1 Gbps	~~1 Gbps.~~
Characteristic	First wire-less comm^n	Digital	Digital, broadband, increased speed	High-speed, all IP	

Technology	Analog cellular	Digital cellular	CDMA, WMTS, TDWT	LTE, Wifi	Wwww

Q.2) Explain GSM architecture?



The Global system for mobile Communication (GSM) architecture is a structured & hierarchical system that enable mobile communication. It consists of various ~~sixty~~ subsystems. It consists of various subsystems, interfaces, & protocols, which allows seamless voice & data transmission. GSM is divided into three primary subsystem:

1. Radio subsystem (RSS):
This subsystem handles the wireless communication between mobile stations & GSM network.
• Mobile Station (MS)
- Includes mobile devices (phone, tablets) & subscriber Identity module (SIM) cards.
- The SIM contains user identity (IMSI), authentication key (Ki) & encryption data.
- Communicates with the Base Transceiver station (BTS) using the Um interface

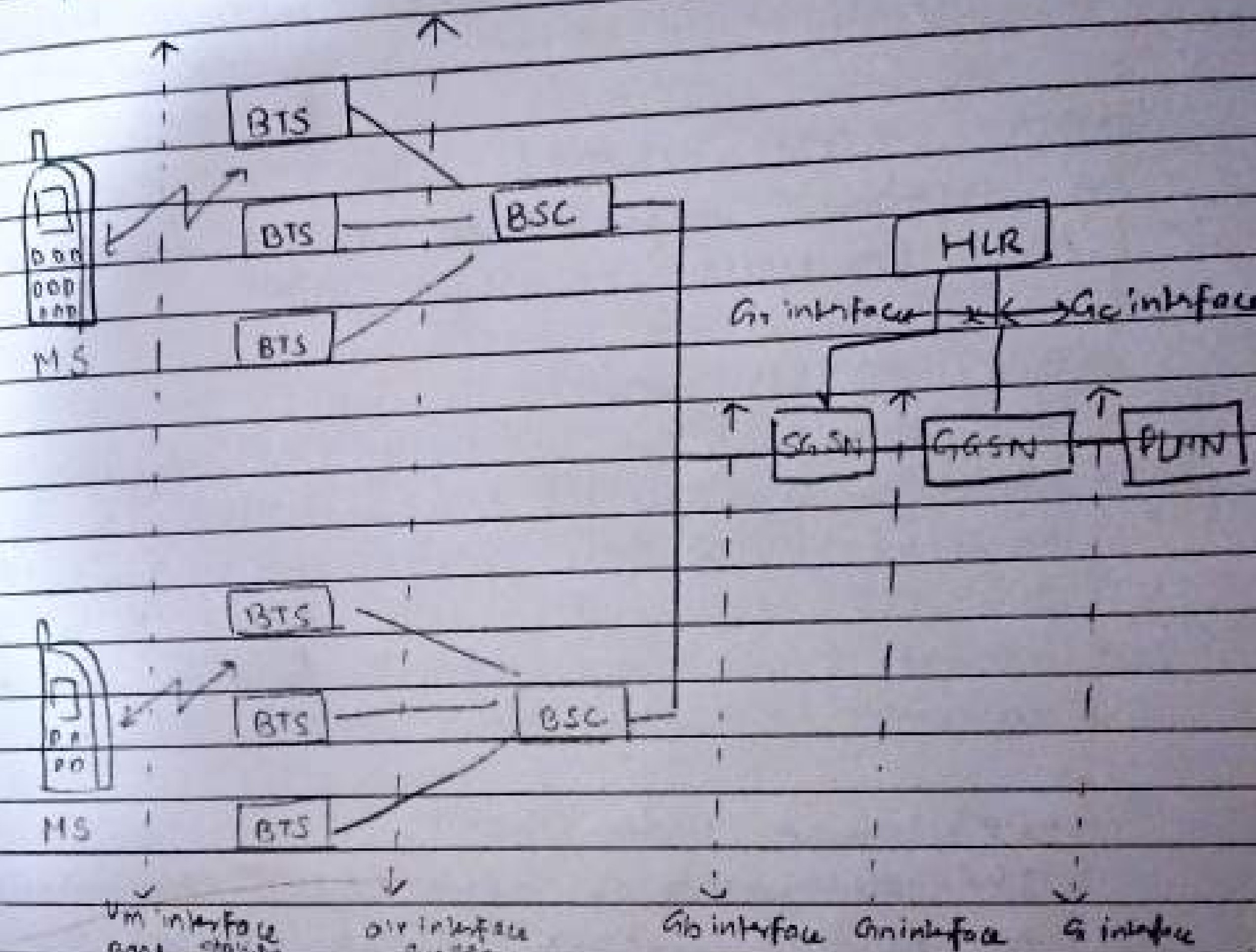
• Base Station Subsystem (BSS):
- Manages radio communication with mobile devices
- Composed of
Base Transceiver station (BTS): Contains antennas Base Station Controller (BSC): Manages multiple BTS, handles frequency allocation, & facilitate handover

2. Network & Switching Subsystems (NSS).
This subsystem manages call routing, mobility, & database ~~services~~ services.
- Uses signalling System No7 (S7) for communication

Interface in GSM.
• Um interface
Between Mobile Station (MS) & Base Transceiver station (BTS).
• Abis Interface:
Between BTS & Mobile switching center (MSC)
• O interface
Between the Operation & Maintenance Centre (OMC) & network elements

GSM architecture is designed for efficient & secure mobile communication ensuring seamless connectivity, mobility management, call switching, making it a widely used standard for 2G mobile networks.

ii) What are the modification require to an existing GSM network to be upgraded to GPRS. Explain with diagram.



General packet radio service (GPRS) is a new bearer service for GSM that greatly improves & simplifies wireless access to packet data networks.

GPRS applies packet radio principal to transfer user data packets in an efficient way between MS & external packet data network.

GSM network element & modification required for GPRS.

• Mobile station (MS) - (GSM network)
New Mobile station is required to access GPRS service. These new terminals will be backward compatiable with GSM voice calls - GPRS

• BTS - (GSM network)
A software upgrade is required in the existing base transreciever site - (GPRS)

• BSC - (GSM network).
The base station controller (BSC) requires a software upgrade & the installation of new hardware called the packet control unit (PCU). The PCU directs the data traffic to the GPRS network & can be a seprate hardware element associated with the BSC - (GPRS).

• GPRS Support Nodes (GSN).
The deployment of GPRS requires the installation of new core network elements called the serving GPRS support node (SGSN) & gateway GPRS support node (GGSN).

• Database (HLR), VLR, etc).
All the databases involved in the network will

require software upgrades to handle the new will nodes & functions introduced by GPRS.

(iii) Explain mobile originated calls (MOC) & mobile terminated call (MTC).

→ Mobile originated call (MoC) & mobile Terminated call (MTC) are two fundamental call process that enable communication between mobile & fixed networks.

1) Mobile Originated Call (MoC).
A MoC is a call that is initiated by a mobile user (Mobile Station - MS) & is directed towards another mobile or landline user.

- Steps involved in MOC:

1. Call involved in MOC MS.
- The mobile user dials the destination number & press on the call button.
- The request is sent via the Um interface to the Base Transreceiver station (BTS).

2. Forwarding to Base Station Controller (BSC).
- The BTS forwards the request to BSC via Abis interface

3. Connection to Mobile switching Centre (MSC).
- The BSC forwards the request to MSC via A interface.
- The MSC checks if user is allowed to make a call (eg. balance check for prepaid users).

4.

4. Routing the call
If the call is allowed, the MSC routes the call
- If the destination is another mobile user, the MSC queries the Home location Register (HLR) & visitor loga location Register (VLR) to find the recipients current location.
- If the destination is a landline user, the call is routed through Public switched Telephone Network (PSTN).

5. Call setup & connection
- The destination network rings the receivers phone
- When the receiver picks up, a two way voice channel is established.

6. Call Termination:
- When either party hangs up, a call release signal is sent.
- The MSC updates billing info^n & release the allocated network resources.

2. Mobile Termination Call (MTC)
A MTC is a call that is recieved by a mobile user (mobile station - MS) from another mobile or landline user.

Steps involved in MTC:
1. Call initiation by a caller.
- A call is placed to a GSM mobile number from another mobile or landline

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2. Call routing via gateway MSC (GMSC)
- The call first reaches GMSC of recipts network.
- The GMSC queries the home location Register (HLR) to find the current visitor location register (VLR) of the recipent.

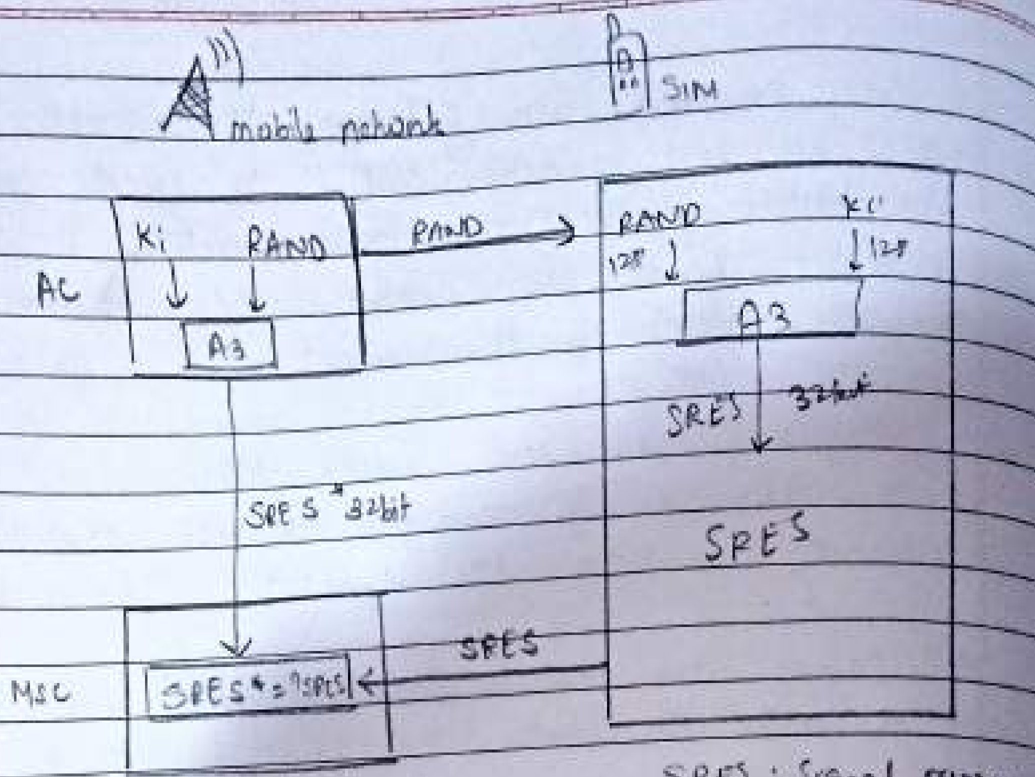
3. forwarding to the serving MSC
- The call is forwarded to MSC that is currently serving the recipent.

4. Paging the Mobile Station (MS).
- The MSC initiates a paging request to locate the recipents phone.
- The BTS in recipents location area (LA) broadcast the paging request.

5. Call ~~set~~ setup & connection.
- When the recipent answers, a voice channel is established between the caller & the recipent

6. Call termination
- When either party hangs up, a call release signal is sent.
- When MSC updates billing records & frees up network resources.

IV) Explain in detail how subscriber Authentication & Encryption is done in GSM.



ki: individual subscriber authen key.
SRES: signal response

Authentication verifies that a user is legitimate before allowing access to the network. It is based on challenge-response mechanism with A3 algorithm.

Steps involved in Authentication:

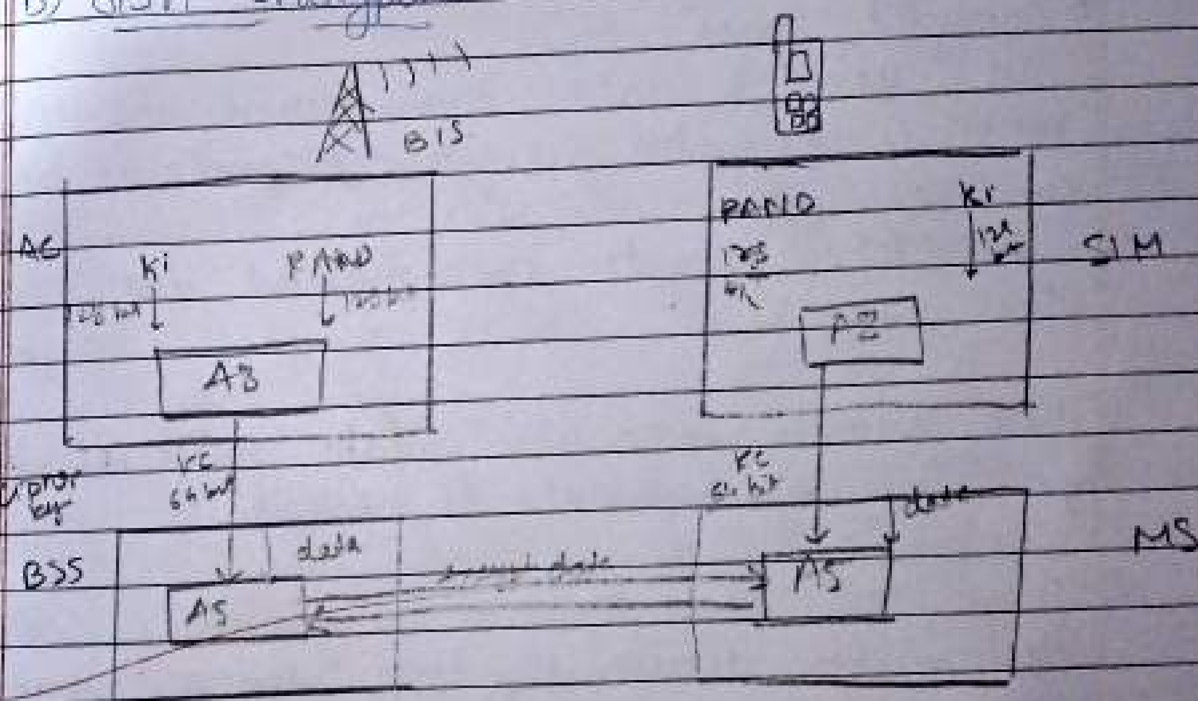
1. Authentication Request
- When a mobile user connects to the network, the MSC requests authentication from AuC.

2. Generating a Random no. (RAND).
- The AuC generates a 128-bit random no (RAND) and it sends it to MS (Mobile station).

3. Computing the signal Response (SRES):
- The SIM card process the RAND using A3 algorithm & the security key (ki) to generate a 32 bit signal Response (SRES)

3. Verification by the Network:
- The MS sends SRES back to network for verification.
- The AuC also computes SRES using the same RAND & Ki
- The MSC compares the recieved SRES with the expected SRES.
• If they match → Authentication succesful
• If they don't match → Access denied

B) GSM Encryption



Once authentication is succesful, GSM activates encryption to secure the commn over the air interface.

Steps to involved in Encryption:
1. Generating the Cipher Key (Kc):
- The SIM & the AuC generate a 64 bit Ciphering key (Kc) using the A8 algorithm

K_c is derived from the RAND & Ki but is never transmitted over the air, ensuring security.

2. Activating Encryption:
- After authentication, the MSC instructs the BTS & MS to enable encryption.
- Both sides were A5 encryption algorithm with K_c to encrypt & decrypt the communication.

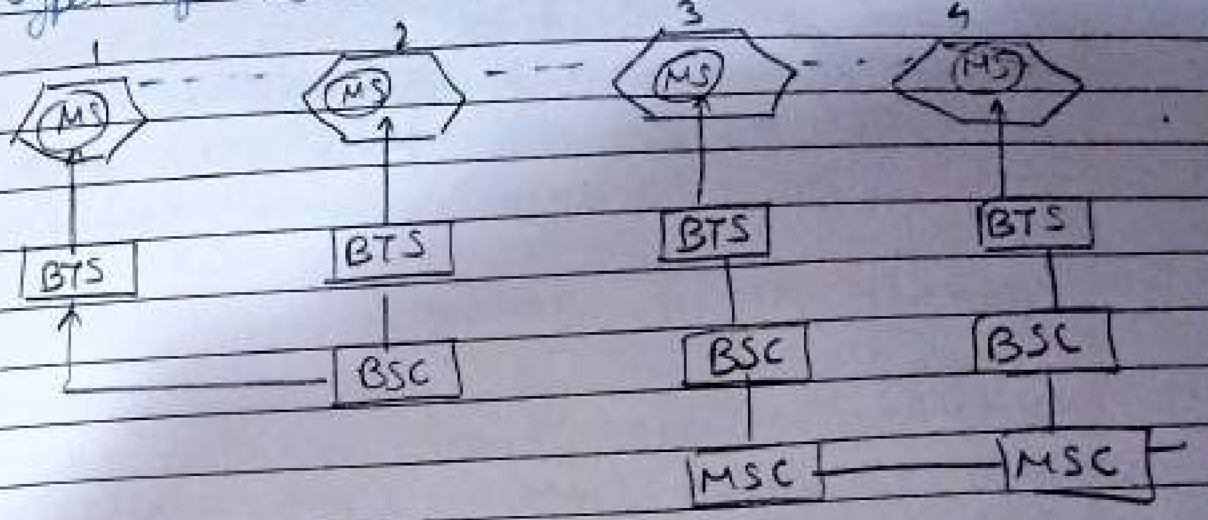
3. Encrypted Comm^n Begins:
- All voice & data transmission b/w MS & BTS is encrypted before transmission over the air interface (Um).
- The BTS decrypts the recieved data using the same K_c.

These security measures make GSM reliable, private & resistant to fraud.

Q) What are the types of handover in GSM? Explain in detail Handover decision making process.

→ In GSM, handover (handoff) refers to the process of transferring an ongoing call or data session from one Base Transceiver station (BTS) or cell to another without call drop. Handover is necessary when a mobile station (MS) moves beyond the coverage area of its current cell or when the system needs to balance network load.

4 Types of Handover



i) Inter cell handover
within a cell, narrow-band interference could make transmission at a certain frequency impossible. The BSC could then decide to change the carrier frequency (scenario 1).

ii) Inter cell, intra-BSC handover:
This is typical handover scenario. The mobile station moves from one cell to another, but stays within the control of the same BSC. The BSC then performs a handover, assigns a new radio channel in new cell & releases the old one (scenario 2).

iii) Inter-BSC, intra MSC handover
As a BSC only controls a limited no of cells, GSM also has to perform handovers between cells controlled by different BSCs. This handover then has to be controlled by the MSC (scenario 3)

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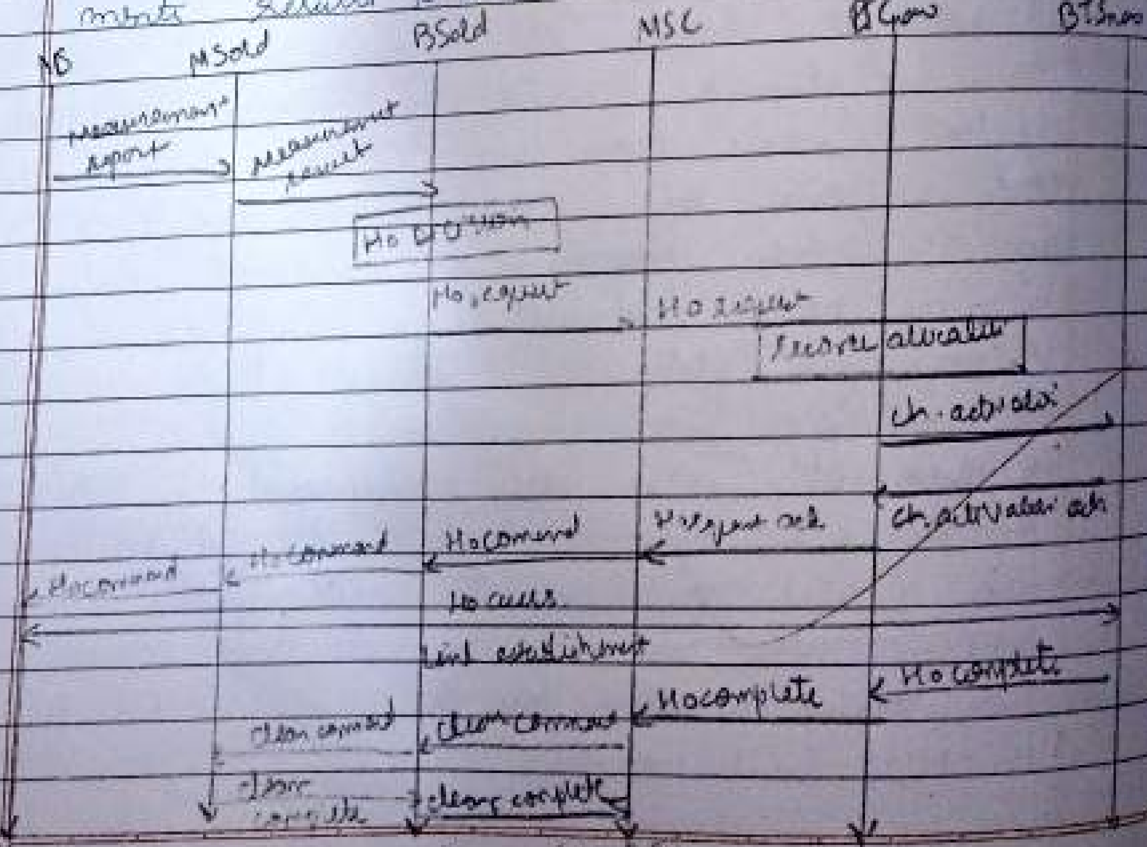
(iv) Inter MSC handover
A handover could be required between two cells belonging to different MSCs. Now both MSCs perform the handover together (scenario 4)

Handover decision making :-

Handover in GSM ensures seamless call transfer when a mobile moves between cells. The decision making process involves multiple steps:

1. Signal Measurement (MS & BTS_old)
- The MS measurements signal strength & reports it to BTS old

2. BTS-old sends data to BSC old
- BTS analyzes the signals & forwards measurements results to BSC



Intra MSC & inter BSC

3. Handover Decision (By BSC old)
- BSCold decides handover based on signal strength, BER, & congestion.
- If needed, it sends a H.O required message to MSC.

4. Resource Allocation (By MSC & BSCnew).
- MSC requests resources from BSC new
- BSCnew allocates resources & activates the new channel.

5 Handover Execution:
- MSC sends H.O command to MS, instructing it to switch to BTS new
- MS connects to BTS new (link Establishment)

6. Completion & Clearing Old Resources:
- H.O complete message confirms a successful transfer
- Old BTS releases resources (Clear Command & clear complete).

This process ensures a smooth transition between BTS, avoiding call drops.

vi) Explain UMTS architecture in detail
→ UTRAN (UTRA Network)
UE (User Equipment)
CN (Core Network)

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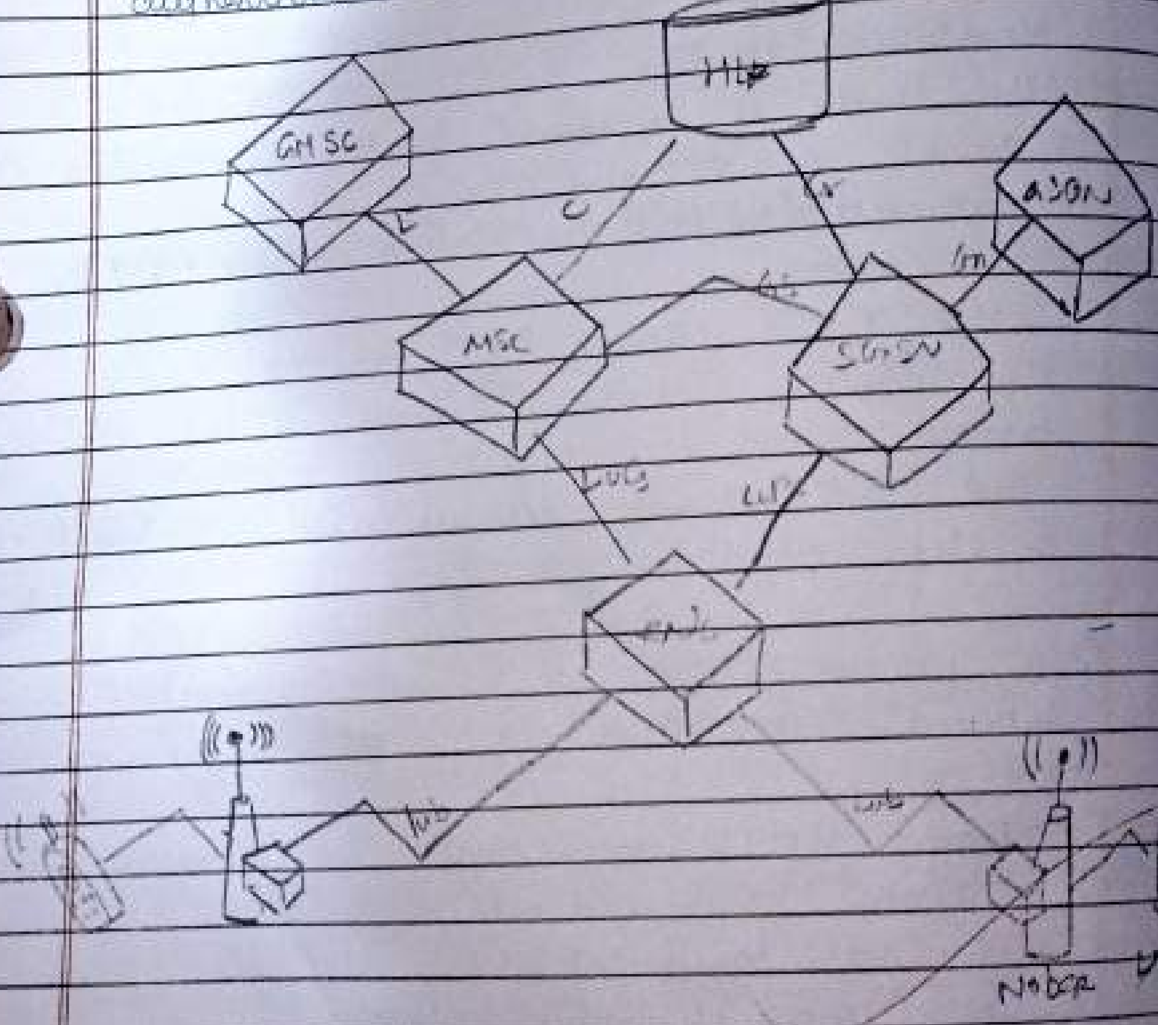
UMTS (Universal Mobile Telecommunication Systems) is a 3G mobile network that improves GSM by enabling higher data rates and multimedia services. It has three main components.

UMTS Architecture:

1. Mobile device with a USIM card for authentication.



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2. UTRAN (UMTS Terrestrial Radio Access Network)
- Handles radio communication:
- Node B (Base stations for transmission
- RNC (Radio Network Controller):
  Manages multiple Node Bs, handover, and resources allocation

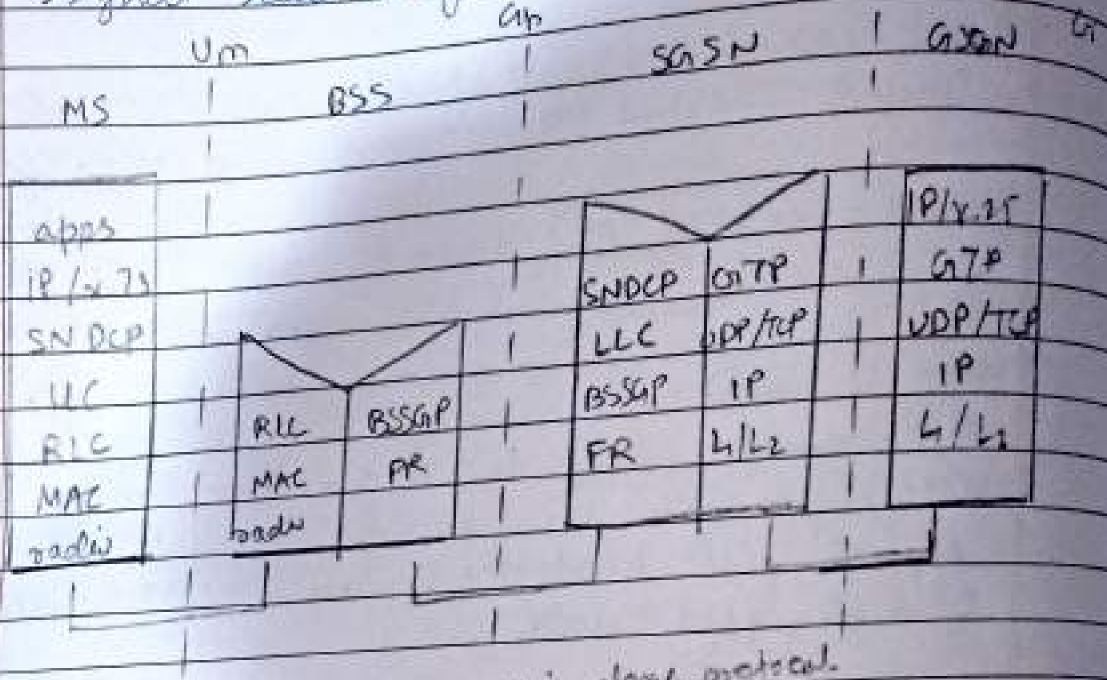
3. Core Network (CN) - Routes calls & data
- Circuit-Switched (CS) domain: Uses MSC for voice calls.
- Packet-Switched (PS) domain: Uses SGSN & GGSN for internet access.

How it works:
- UE connects to Node B → RNC → Core Network.
PSTN (for voice) or Internet (for data).
- supports fast handover, multimedia applications, & high speed data.

vii) Explain GPRS protocol stack Architecture in detail.

→ GPRS (General Packet Radio Service) is an extension of GSM that enables packet-switched data transfer for internet access, multimedia messaging, & email services. It introduces a new protocol stack optimized for data communication.
GPRS protocol stack (layer wise breakdown).

A) Physical layer (Air Interface).
Handles radio transmission between MS (Mobile Station) & BTS.
- Uses GSM for time division multiple access (TDMA) for sending data in bursts.
- frequency hopping & error correction ensure signal reliability.



GPRS transmission plane protocol architecture

Network layer (packet-switched comm^n).
- It is layer is divided into two sub-layers:
Sub-Network dependant Coverage Protocol (SNDCP)
- Compress IP header to reduce data.
- Multiplexes multiple network layer connections into one logical link.

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ii) logical link Control (LLC) layer
- Ensure secure communication between the MS & SGSN.
- provides encryption, error correction, & retransmission.

iii) Base station system GRPS Protocol (BSSGP)
- Handles routing & QoS (Quality of Service) between BTS & SGSN.
- No error recovery (relies on LLC for reliability)

iv) Network service (NS) layer.
- Uses frame Relay protocol to transfer data b/w BSC & SGSN.

c) Application layer (Higher-level Protocols).
This layer supports end-user applications such as:
- IP (Internet Protocol) used for web-browsing & data transfer.
- X.25 protocol - for legacy-applications requiring packet-switching networks.
- WAP (wireless Application Protocol) - Optimized for mobile internet browsing.

* End to end Data flow in GPRS.
1. MS sends data → SNDCP compresses → LLC encrypts → BSSGP routes.
2. SGSN process mobility & security → GGSN forwards to internet

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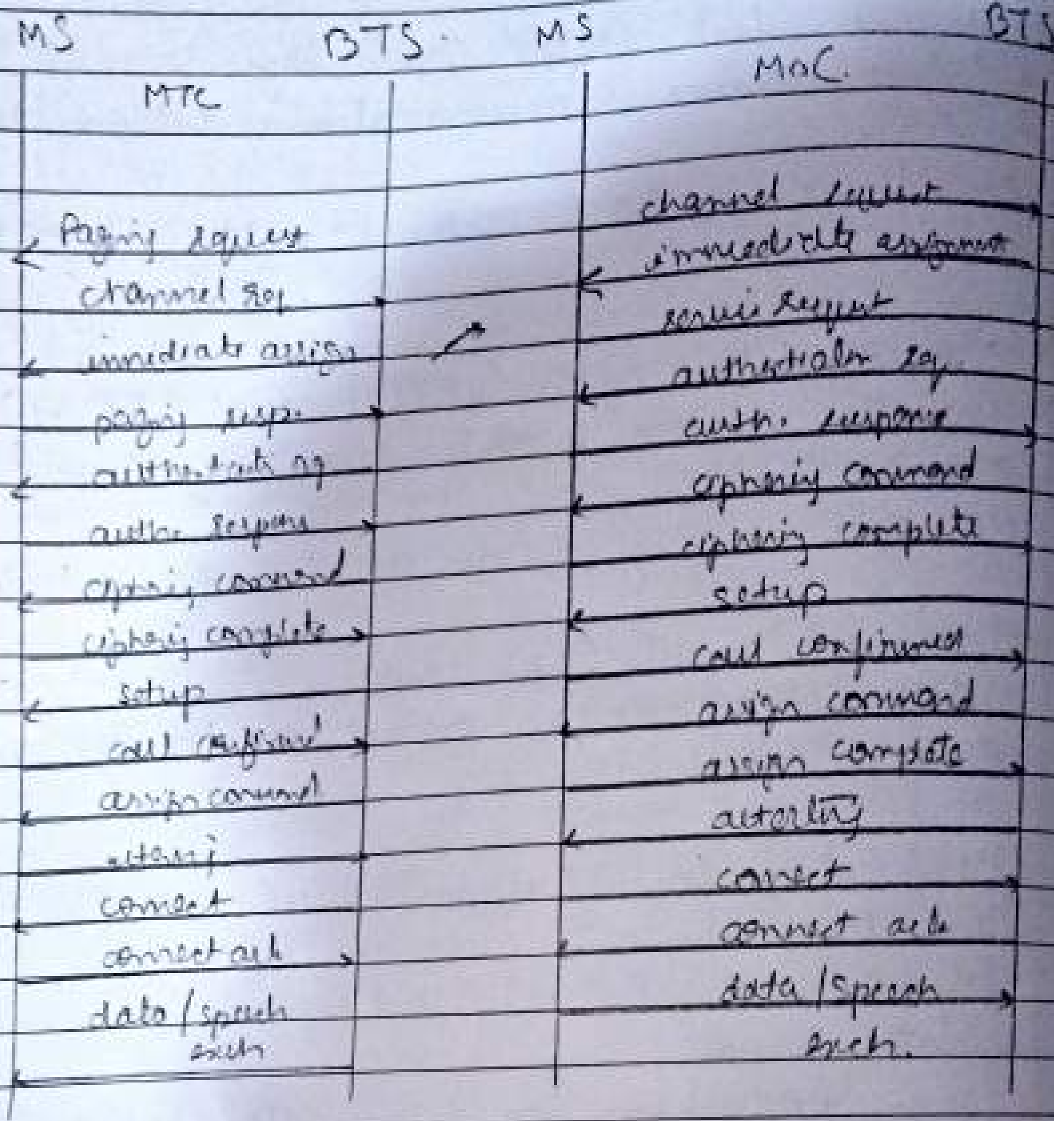

Message flows for MTC & MOC

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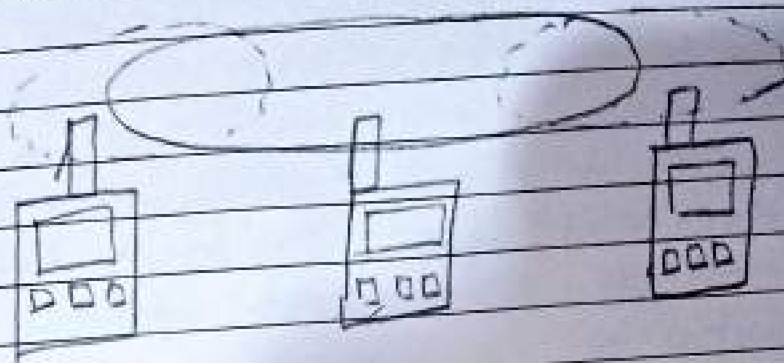
Q.3] i) Explain hidden station & exposed station problems in WLAN.

→



Wireless LANs (WLANs) face hidden station & exposed station problems due to the nature of wireless commⁿ. These b issues impact the carrier sense multiple Access (CSMA) protocol, leading to collisions or unnecessary transmission delays

1. Hidden Station Problem.

A hidden station occurs when two devices cannot detect each others transmission due to their limited transmission range, but both are trying to communicate with a common receiver

example scenario:

- Station A & station C want to send data to station B.
- A & C cannot detect each other, so C as assumes the channel free.
- Both A & C transmit simultaneously, causing a collision at B.
- Neither A nor C can detect the collision, leading to packet loss.

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Impact:

· Increases packet loss, reduces network

· Causes idle time in wireless comm

· Solution:

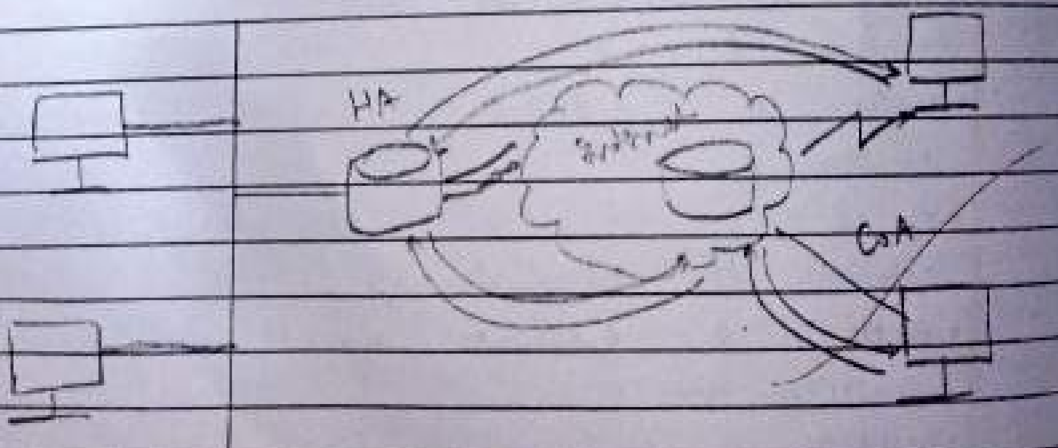
· MACA protocol.

- B sends RTS to A, & A reply with CTS

- C, C does not hear CTS, so it knows A is outside its range & can proceed with its transmission

ii) Why is mobile IP packet required to be forwarded through a tunnel. Explain minimal techniques of encapsulation of mobile IP packet

→ ① In mobile IP, a mobile node (MN) moves from its home network to a foreign network while maintaining a permanent home IP address. However, traditional IP routing is based on fixed address, so packets destined from the IP address will be sent to the home network instead of current location of the MN.



Mobile IP

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Encapsulation is the process of adding an outer header to a packet to forward it through a tunnel. Minimal encapsulation reduces overhead by avoiding unnecessary headers.
Minimal encapsulation results in less overhead & can be used if the mobile node, home agent & foreign agent all agree to do so.
With minimal encapsulation, the new header is inserted between the original IP header & the original IP payload. It includes the following fields.

· Protocol: Copied from the destination address field in the original IP header. This field identifies the protocol & thus identifies the type of headers that begins the original IP payload.

· S: If 0, the original source address is not present, & the length of this header is 8 octets.

· If 1 original source address is present & the length of this header is 12 octet.

· Header checksum: Computed over all the fields of this location.

· Original destination address: Copied from the destination add field in original IP header

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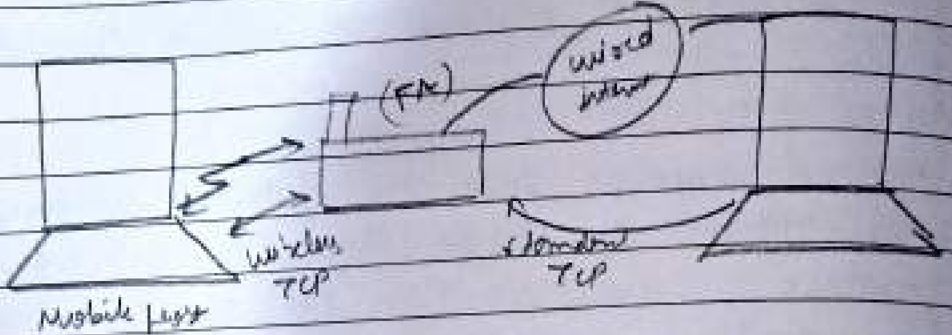
(Ti) Explain different types of mobile TCP in detail
→ Mobile TCP (M-TCP) is a set of modifications to the traditional TCP to improve performance in wireless & mobile env. Since in wireless & ~~mob~~ standard TCP was designed for wired networks, it struggles with high packet loss, handover, & disconnections in mobile networks. Various Mobile TCP var's help address these issues



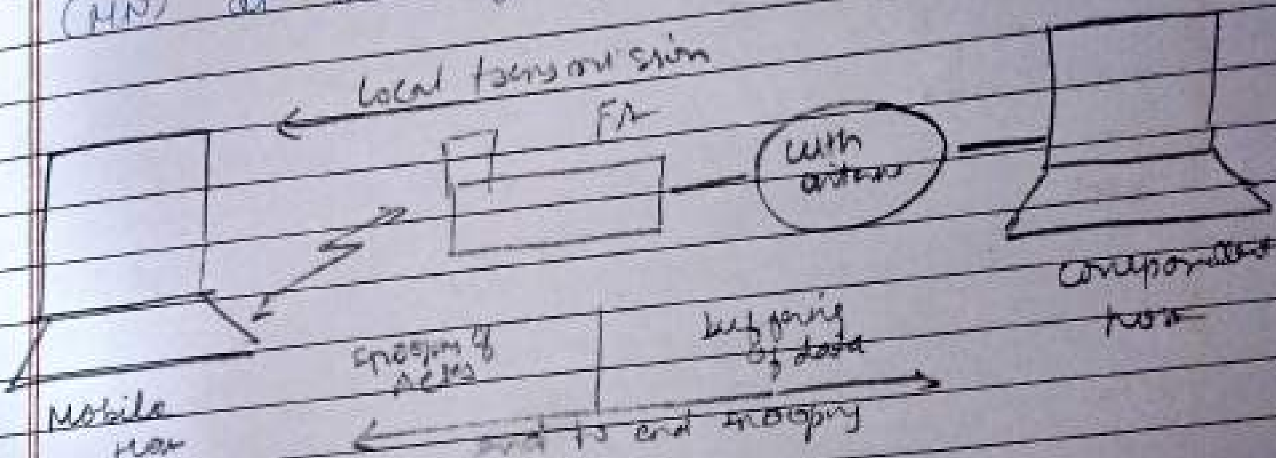
. I-TCP splits the TCP connection into two parts.
1. Fixed TCP connection (B/w correspondent host (CH) & foreign agent (FA))
2 wireless TCP connection (B/w FA & mobile phone (MH))

How it work
. The FA acts as a proxy & terminates the TCP connection from the CH.
. The FA then establishes a new TCP connection with the MH using a wireless-optimized TCP
. If packets are lost on the wireless link, the FA retransmits locally, preventing packet loss from affecting the CH.

Advantages:
- fast local retransmission
- No changes needed in correspondent host

Disadvantages:
- Breaks end to end TCP semantics
- loss of buffered data if FA crashes

2. Snooping TCP:
Snooping TCP monitors TCP traffic between the corresponding host (CH) & Mobile Node (MN) at the foreign Agent (FA)



How it works:
- The FA "snoops" (monitors) TCP packets & acks unacknowledged packets.
- If a packet is lost on the wireless link, the FA retransmits locally, avoiding retransmission from the CH.
- The end to end TCP connection remains intact as the CH is unaware of retransmission.

Advantages:
- Preserves end to end TCP semantics

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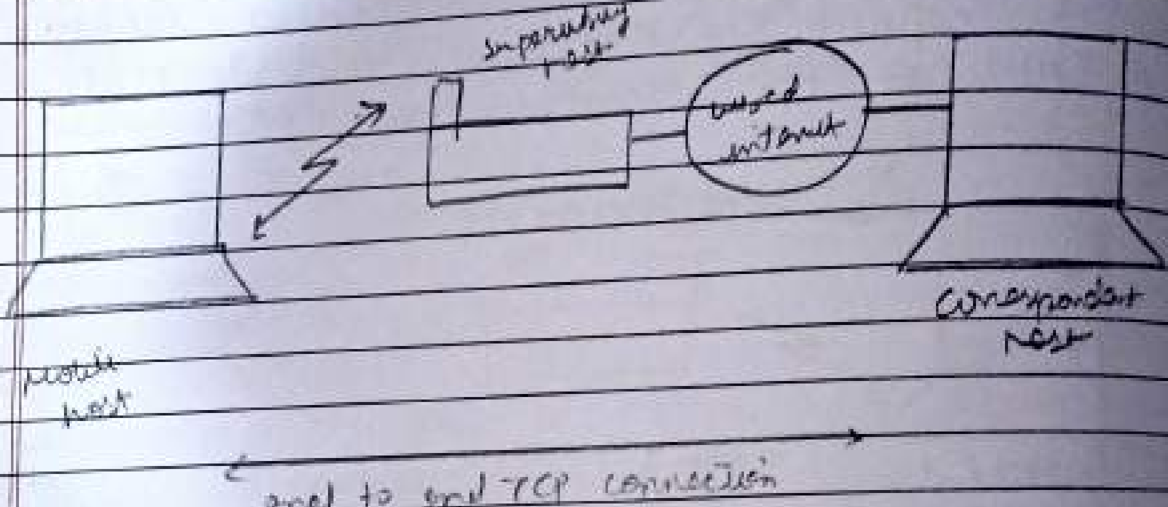
• Faster local retransmission

Disadvantages
• Does not work with encrypted TCP
• Increases processing overhead.

3. Mobile TCP (M-TCP)
M-TCP is designed to handle for disconnection & handovers in mobile environments



How it works.
• A supervisory host (SH) monitors the connection between the CH & MH.
• When a disconnection to zero, prevents unnecessary retransmission.
• Once the MH reconnects, the SH restores the TCP window, allowing normal transmission to resume.

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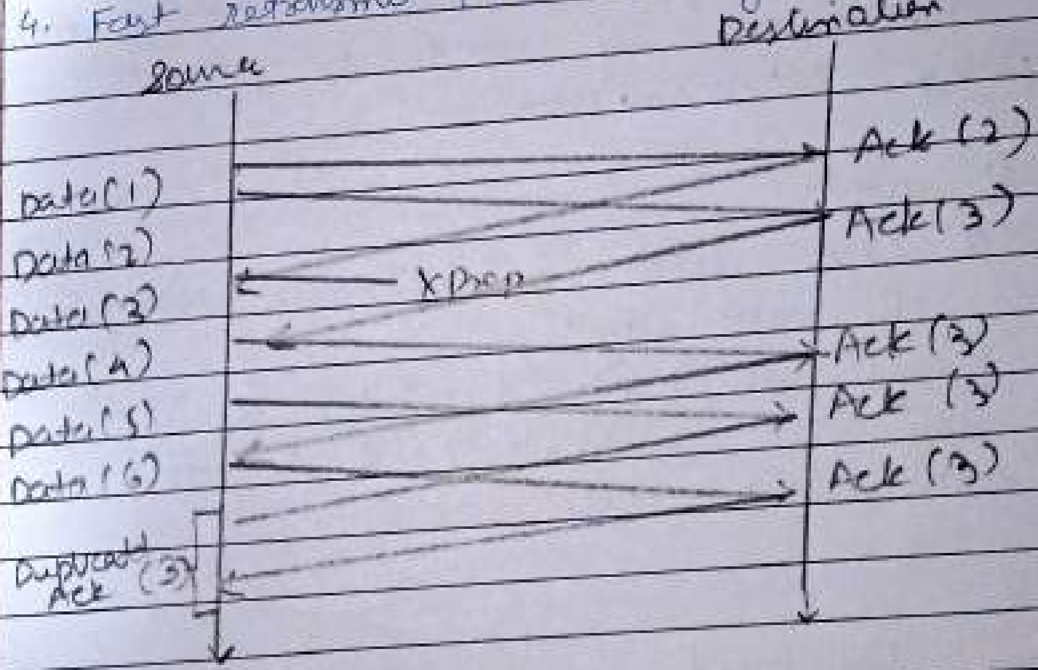
Advantages:-
· Prevents unnecessary retransmission
· Maintains TCP state during disconnection.

Disadvantages:
· Requires additional supervisory host (SH)
· Packet loss still affects sender.

4. Fast retransmit (Fast recovery TCP.



This technique prevents TCP from going into slow start when a packet is lost due to mobility

How it works
- If a handoff occurs, the MN sends duplicate acks to the CH.
- The CH then immediately retransmits lost packets instead of reducing it congestion window

Advantages.
- Avoids slow start after handover.
- Improves data transfer speed.

Disadvantage.
- Not effective in high loss environment

5. Timeout freezing TCP:
This technique pauses TCP timers when a mobile node moves out of range to prevent unnecessary retransmission

How it works:
- The MAC layer detects an upcoming disconnection & informs the TCP layer
- TCP freezes its timer & stops sending packets
- Once the MN reconnects, TCP resumes transmission from the last state

(iv) Explain the need of specialized MAC in wireless communication

→ Need for specialized MAC in wireless communication:
The Media Access Control (MAC) layer in wireless communication manages how devices access & share the wireless medium efficiently. Unlike wired networks, where Carrier Sense Multiple Access with Collision Detection (CSMA/CD) works well, wireless networks face unique challenges that require a specialized MAC protocol.

C-5-4

Challenges in wireless Communication:

1. signal length decrease with distance.
2. Hidden Terminal Problem
3. Exposed Terminal Problem.
4. mobility of nodes
5. Lack of Central Coordinator.
6. Half duplex Transmission

21.2.23